

Lecture 13: Related rates

Motivation: sometimes there are related quantities that are changing together in time. If we know the rate of change of one of these quantities, can we find the rate of change of the other?

Example: air is being pumped into a spherical balloon

so that its volume increases at a rate of $100 \text{ cm}^3/\text{s}$ $\frac{dV}{dt} = 100$

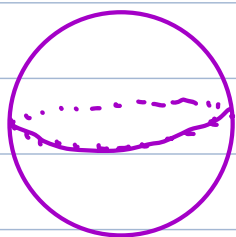
How fast is the radius of the balloon increasing if it is currently 25 cm ? $\frac{dr}{dt}$

Solution:

quantity whose rate of change we know: volume = $V(t)$

quantity whose rate of change we want to find: radius = $r(t)$

Idea: write an equation relating these two quantities



$$V = \frac{4}{3} \pi r^3 \quad (\text{volume of sphere})$$

$$\frac{d}{dt} \left(V(t) = \frac{4}{3} \pi (r(t))^3 \right)$$

$$\frac{dV}{dt} = \frac{4}{3} \pi \cdot \underbrace{\frac{d}{dt} (r(t))^3}_{\text{chain rule: } 3r^2 \cdot \frac{dr}{dt}}$$

$$\text{chain rule: } 3r^2 \cdot \frac{dr}{dt}$$

$$100 \frac{dv}{dt} = \frac{4}{3} \pi \cdot r^2 \frac{dr}{dt} = 4\pi \frac{r^2}{25} \frac{dr}{dt}$$

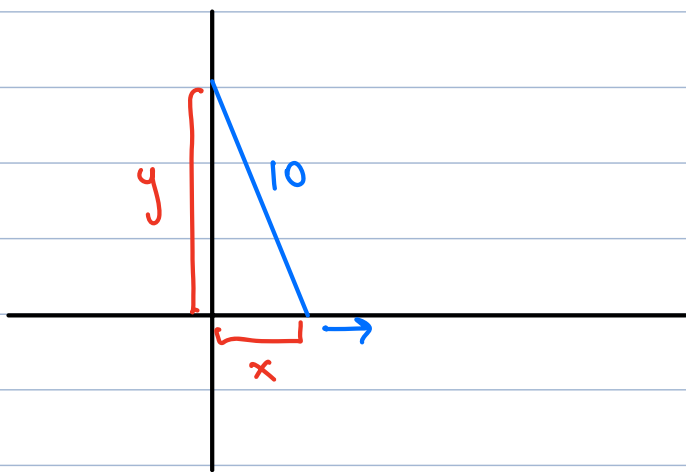
Now use given information ($\frac{dv}{dt} = 100$, $r = 25$)

$$100 = 4\pi \cdot (25)^2 \cdot \left(\frac{dr}{dt}\right) \Rightarrow \frac{dr}{dt} = \frac{100}{4\pi \cdot (25)^2} = \frac{1}{25\pi} \frac{\text{cm}}{\text{s}}$$

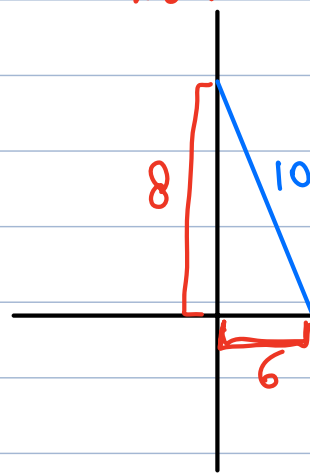
Example: A ladder 10 ft long rests along a vertical wall.

If the bottom of the ladder slides away from the wall at a rate of

1 ft/sec, how fast is the top of the ladder sliding down the wall when the bottom of the ladder is 6 ft from the wall?



"Snapshot" at current moment in time: $x = 6$



Known rate: x

unknown rate: y

Find equation relating x and y :

Pythagoras: $x^2 + y^2 = 10^2$

$$\begin{aligned} 6^2 + y^2 &= 10^2 \\ y^2 &= 100 - 36 \\ &= 64 \\ y &= 8 \end{aligned}$$

$$(x(t))^2 + (y(t))^2 = 100 \quad \frac{d}{dt}$$

$$2x \cdot \frac{dx}{dt} + 2y \cdot \frac{dy}{dt} = 0$$

Use given information:

$$\frac{dx}{dt} = 1, \quad x = 6 :$$

$$2 \cdot 6 \cdot 1 + 2y \cdot \frac{dy}{dt} = 0$$

$$y = 8 :$$

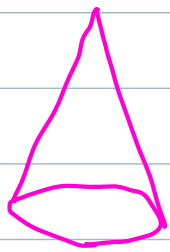
$$12 + 16 \cdot \frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{12}{16}$$

$$= -\frac{3}{4} \frac{\text{ft}}{\text{sec}}$$

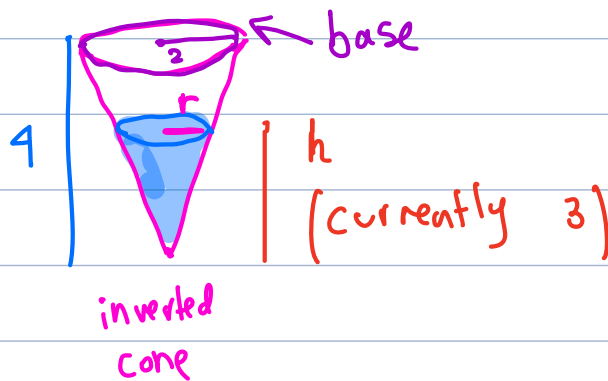
how fast is the top of the ladder sliding down the wall :

$$\frac{3}{4} \frac{\text{ft}}{\text{sec}}$$

Example: A water tank has the shape of an inverted circular cone with base radius 2 m and height 4 m . If water is being pumped into the tank at a rate of $2\text{ m}^3/\text{min}$, find the rate at which the water level is rising when the water is 3 m deep.



cone



inverted cone

quantity whose rate we know? V

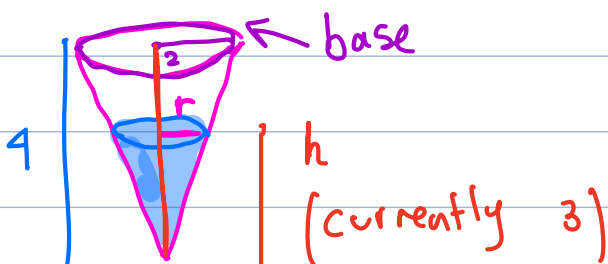
quantity whose rate we want to find? h

Find equation relating V and h :

Volume of cone: $V = \frac{1}{3} \cdot (\underbrace{\text{area of base}}_{\pi r^2}) \cdot \underbrace{\text{height}}_h$

$$V = \frac{1}{3} \pi r^2 h$$

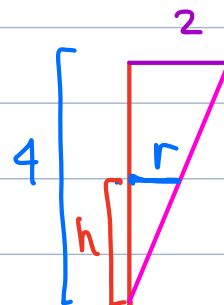
Question: how to find r ?



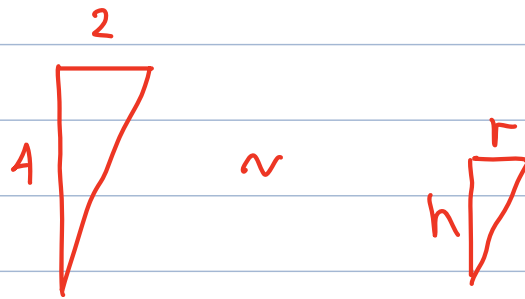
inverted cone

profile view

$\Rightarrow$



Similar triangles!



$$\frac{r}{h} = \frac{2}{4} = \frac{1}{2} \Rightarrow \boxed{r = \frac{h}{2}}$$

$$V = \frac{1}{3} \pi r^2 h$$

$$V = \frac{1}{3} \pi \left(\frac{h}{2} \right)^2 h = \frac{1}{3} \pi \cdot \frac{h^2}{4} \cdot h = \boxed{\frac{\pi}{12} h^3 = V}$$

$\frac{d}{dt}$

$$\frac{\pi}{12} \cdot 3h^2 \cdot \frac{dh}{dt} = \frac{dV}{dt}$$

plug in given information:

$$\bullet \frac{dV}{dt} = 2$$

$\Rightarrow$

$$\frac{\pi}{12} \cdot 3 \cdot 3^2 \cdot \frac{dh}{dt} = 2 \Rightarrow$$

$$\bullet h = 3$$

$$\boxed{\frac{dh}{dt} = \frac{8}{9\pi} \frac{\text{m}}{\text{min}}}$$

Summary of general method/strategy

Step 0: Preparations for solving.

Read the problem carefully. Draw a diagram to illustrate and give symbols to each quantity that depends on time. Express the given information and the rate we want to find in terms of derivatives.

Step 1: Write an equation that relates quantities whose rate we know to quantity whose rate we want to find

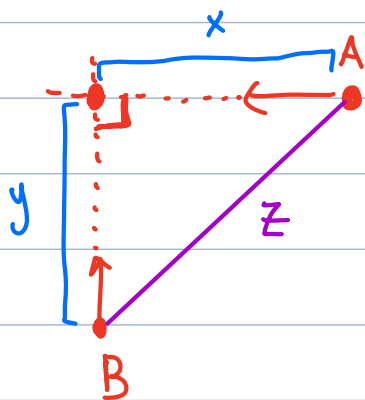
If necessary, use the geometry of the situation to eliminate one of the variables by substitution.

Step 2: Use the chain rule to differentiate both sides of the equation with respect to t .

Step 3: Substitute the given information into the resulting equation and solve for the unknown rate.

Example: Car A is traveling west at 50 mi/h and Car B is traveling north at 60 mi/h. Both are headed for the intersection of the two roads. At what rate are the cars approaching each other when car A is 0.3 mi and car B is 0.4 mi from the intersection?

Step 0



given: $\frac{dx}{dt} = -50 \frac{\text{mi}}{\text{h}}$

$\frac{dy}{dt} = -60 \frac{\text{mi}}{\text{h}}$

unknown rate: $\frac{dz}{dt} = ?$

Step 1

equation relating our variables: $x^2 + y^2 = z^2$

Step 2: $\frac{d}{dt}(x^2 + y^2) = \frac{d}{dt}(z^2)$

$$2x \cdot \frac{dx}{dt} + 2y \cdot \frac{dy}{dt} = 2z \cdot \frac{dz}{dt}$$

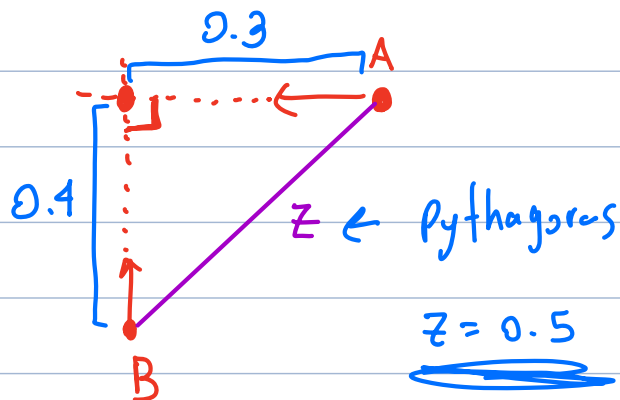
Step 3: plug in knowns.

$$\frac{dx}{dt} = -50$$

$$x = 0.3$$

$$\frac{dy}{dt} = -60$$

$$y = 0.4$$



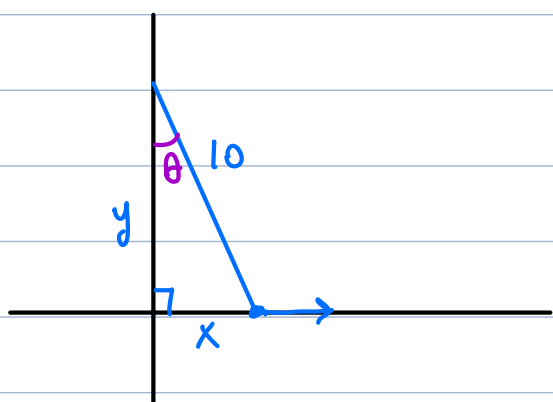
$$\underbrace{2 \cdot (0.3) \cdot (-50)}_{-30} + \underbrace{2 \cdot (0.4) \cdot (-60)}_{-48} = \underbrace{2 \cdot (0.5)}_1 \cdot \frac{dz}{dt}$$

$$\boxed{\frac{dz}{dt} = -78 \frac{\text{mi}}{\text{h}}}$$

Example: A ladder 10 ft long rests along a vertical wall. If the bottom of the ladder slides away from the wall at a rate of

1 ft/sec, how fast is the angle between the top of the ladder and the wall changing

when the bottom of the ladder is 6 ft from the wall?



Known rate: $\frac{dx}{dt} = 1 \frac{\text{ft}}{\text{sec}}$

rate we want: $\frac{d\theta}{dt} = ?$

Want an equation relating known and unknown rates

x and θ

~~$x^2 + y^2 = 10^2$~~

$\sin(\theta) = \frac{\text{opp}}{\text{hyp}} = \frac{x}{10}$

$\frac{d}{dt} \left(\sin(\theta) \right) = \frac{d}{dt} \left(\frac{x}{10} \right)$

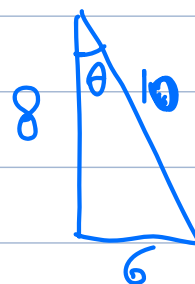
$\cos(\theta) \cdot \frac{d\theta}{dt} = \frac{1}{10} \cdot \frac{dx}{dt} \quad (*)$

use our givens to find $\frac{d\theta}{dt}$:

$\frac{dx}{dt} = 1, \quad \cos(\theta) = \frac{\text{adj}}{\text{hyp}} = \frac{8}{10}$

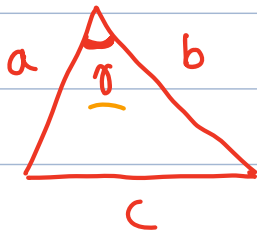
$\star \quad \frac{8}{10} \cdot \frac{d\theta}{dt} = \frac{1}{10} \cdot 1 \Rightarrow$

$\frac{d\theta}{dt} = \frac{1}{8} \frac{\text{radians}}{\text{sec}}$



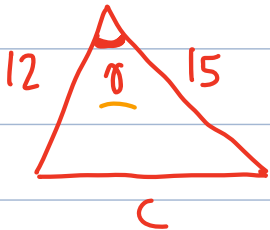
Example Two sides of a triangle have lengths 12 m and 15 m. The angle between them is increasing at a rate of $2^\circ/\text{min}$. How fast is the length of the third side increasing when the angle between the sides of fixed length is 60° ?

Hint: use law of cosines



$$c^2 = a^2 + b^2 - 2ab \cos \theta$$

Step 0: make a diagram, identify known and unknown rates



$$c^2 = \underbrace{12^2 + 15^2}_{360} - \underbrace{2 \cdot 12 \cdot 15 \cdot \cos \theta}_{360}$$

known rate: $\frac{d\theta}{dt} = \frac{2^\circ}{\text{min}} = \frac{2^\circ \cdot \pi \text{ radians}}{180^\circ \text{ min}} = \underline{\underline{\frac{\pi}{90} \frac{\text{radians}}{\text{min}}}}$

unknown rate: $\frac{dc}{dt} = ?$

Step 1: Find an equation relating quantities of known and unknown rates.

$$c^2 = \underbrace{12^2 + 15^2}_{144 + 225} - \underbrace{2 \cdot 12 \cdot 15 \cdot \cos \theta}_{360}$$

" 369

$$c^2 = 369 - 360 \cos \theta$$

Step 2: Differentiate both sides with respect to t

$$\frac{d}{dt}(c^2) = \frac{d}{dt}(369 - 360 \cos \theta)$$

$$2c \cdot \frac{dc}{dt} = -360 \cdot (-\sin \theta) \cdot \frac{d\theta}{dt} \quad (*)$$

Step 3: plug in known info

$$\underline{\frac{d\theta}{dt} = \frac{\pi}{90}}, \quad \theta = 60^\circ = \frac{\pi}{3}, \quad c = ?$$

$$c^2 = 369 - 360 \cos \theta$$

$$\text{at current moment: } \theta = 60^\circ \Rightarrow \cos \theta = \frac{1}{2}, \quad \left(\sin \theta = \frac{\sqrt{3}}{2} \right)$$

$$\Rightarrow c^2 = 369 - \frac{360}{2} = 369 - 180 = 189$$

$$c = \sqrt{189}$$

$$2c \cdot \frac{dc}{dt} = -360 \cdot (-\sin \theta) \cdot \frac{d\theta}{dt} \quad (*)$$

$$\Rightarrow 2 \cdot \sqrt{189} \frac{dc}{dt} = 360 \cdot \left(\frac{\sqrt{3}}{2} \right) \cdot \frac{\pi}{90}$$

$$2 \cdot \sqrt{189} \frac{dc}{dt} = 2\sqrt{3} \cdot \pi \Rightarrow$$

$$\frac{dc}{dt} = \frac{-\sqrt{3} \pi}{\sqrt{189}} =$$

$$\boxed{\frac{\pi}{\sqrt{63}} \frac{m}{min}}$$